Discrete Structures

Part 0: Introduction & Course Organization

Ivan Bliznets

University of Groningen

Overview

1. Welcome
2. Motivation
3. Structure

Outline

1 Welcome
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Welcome to Discrete Structures!



Ivan Bliznets
lecturer, course coordinator

Teaching Assistants

Tutorials will be given by:

- Bent Goelema
- Stefan van der Bij
- Teodor Pintilie

Outline

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Discrete (Mathematical) structures

Discrete: refers to things that are separate from each other and not connected. For example, in mathematics, discrete variables are those that have distinct and separate values, as opposed to continuous variables which can take on any value within a range. Calculus deals with continuos variables.

Mathematical: rigorous, precise, unambiguous.

Structures: the way something is composed of parts, or organized.

Motivation

Studying discrete structures is crucial for future software developers for several reasons:

- Foundation of Computer Science
- Problem-Solving Skills
- Algorithm Design and Analysis
- Graph Theory and Networks
- Logic and Formal Verification
- Database Theory
- Career Opportunities

Learning Objectives (from Ocasys)

At the end of the course, you are able to:

1. Apply definitions and constructions pertaining to discrete mathematical objects such as numbers, recurrences, sets, functions, relations, partially ordered sets, graphs, and trees.
2. Count discrete mathematical objects such as sets and sequences.
3. Carry out simple mathematical proofs, including proofs by (strong) mathematical induction, to formally justify properties of discrete mathematical structures.
4. Apply various algorithms for solving problems or computing properties of relations, trees and graphs.

Related Courses – BSc Computing Science

- Introduction to Logic (Y1 - Ia) ✓
- **Discrete Structures (Y1 - Ib)**
- Programming Fundamentals (Y1 - I)
- Algorithms and Data Structures in C (Y1 - IIa)
- Advanced Algorithms (Y2 - Ia)
- Languages and Machines (Y2 - IIb)
- Computational Complexity (Y3 - Ia)

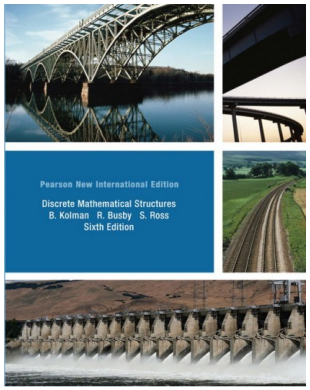
Outline

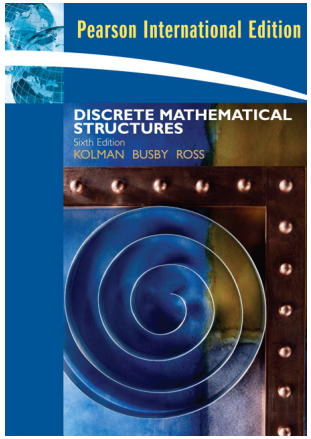
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Textbook and Slides (and their role)





1. B. Kolman, R. Busby, S. C. Ross. Discrete Mathematical Structures. 6th edition (Pearson New International Edition).
2. Oscar Levin. Discrete Mathematics. https://discrete.openmathbooks.org

- **Textbooks:** main reference: weekly readings and suggested tutorial exercises.
- **Slides:** main support during lectures. Published in Brightspace (under Lectures). Sometimes updates/corrections after the lecture.
- Some (small) parts of the book have to be studied on your own.
- Both textbook readings and slides are examination material.

Course Overview

Book	Topic
1.1 - 1.6	Sets, sequences, languages, matrices, structures
2.3 - 2.4	Proofs and mathematical induction
3.1 - 3.3	Permutations, combinations, pigeonhole principle
3.5	Recurrence relations
4.1 - 4.5	Partitions, relations, digraphs, equivalences
4.7 - 4.8	Operations on relations, transitive closure
5.1 - 5.4	Functions
6.1 - 6.2	Partially ordered sets, extremal elements
6.3	Lattices
6.4 - 6.5	Boolean algebras
7.1 - 7.5	Trees
7.1 - 7.5,	8.1 - 8.3 (Kolman et al.)
Chapter 2 (Levin)	Graph theory

Note: Sections 2.1-2.2 (Logic) are assumed to be known and are not explicitly addressed. Consider them part of the exam material. Study them if you have not taken the logic course.

Course setup

All information on course organisation is found in Brightspace.
Make sure to read it all! Only a few remarks here.

5 ECTS: approximately 13 hours per week + 36 hours exam preparation.

- **Lectures (4h):** Wed 11-13h and Wed 13-15h
- **Tutorials (4h):** see rooster.rug.nl
  - Prepare for tutorials by doing/looking at suggested exercises in advance. Know which ones you want to discuss in details.
  - Some additional exercises and quizzes posted in Brightspace.
- **Self study (5h):** reading, exercises, homeworks, quizzes.

Grade

- Quizzes, homeworks, active participation gives you 10% of your grade (plagiarism is pointless).
- Exam 90% of your final grade
- You can take resit only if you did not pass the course
- Resit 90% of your final grade
- **Exam, resit:** closed-book, mostly pen-paper

Any Questions?